

Multi-image statistical analysis for indoor optical camera communication considering spatial combining and practical receiver orientations

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Images are essential for capturing transmitted data in Optical Camera Communication (OCC) systems. Yet, most studies focus on single, independent images, overlooking the benefits of multi-image approaches for spatial combining in OCC. Additionally, the statistical effects of practical camera orientations remain underexplored, as most assume a fixed and vertically upward receiver. This Letter addresses these gaps by investigating the potential of multi-image processing in indoor OCC through two cases. First, we introduce three image-combining techniques to evaluate OCC performance across various phone orientations and user positions using consecutive images captured in a repetitive transmission scheme. Second, based on measured phone orientation data, we analyze the statistical impact of practical phone use-induced orientations on imaging location and intensity. We further examine the capacity for multiple independent LED transmissions at different positions. Our research aims to unlock the potential of multi-image processing through spatial combining techniques, providing a new statistical perspective on OCC performance analysis with practical receiver orientations involved.

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1. INTRODUCTION

Optical Camera Communication (OCC), a subset of Optical Wireless Communication (OWC), repurposes light and uses cameras to transform physical spaces into digital portals for location-aware and identity-based communication. As a complement to Radio Frequency (RF)-based solutions, OCC shows great promise in applications like Smart Homes and Intelligent Transportation Systems [1].

Images are essential for extracting transmitted information in camera-assisted OCC systems. Most studies focus on analyzing individual images from the acquired dataset independently for OCC system modeling and performance analysis. Methods for individual image generation considering both parallel and angular distortions at the transceiver were proposed in [2] and [3]. BER analysis of OCC exposure effects was presented in [4] which treated each image as a separate study object. However, limited research has treated multiple consecutive images as a unified study object. The frame difference method integrates consecutive images for transmitter tracking. A 30-frame Frame-Averaging Background Removal (FABR) technique was introduced in [5]. High-order statistics for interframe analysis in [6] were also explored for OCC systems. Among them, the potential of multiple images in spatial combining is rarely involved. Besides, the camera receiver is always assumed to be vertically upward and fixed in most investigations while their statistical impact is insufficiently explored.

Motivated by the above, in this Letter, we provide a statistical indoor OCC performance analysis based on multiple images where 2×2 Light-Emitting Diodes (LEDs) on the room ceiling and users' handheld mobile phone cameras are transmitters and receivers respectively. Two typical cases are considered, according to whether the multiple images investigated are obtained after repetitive transmissions. In the first case, three typical image-driven combining techniques including Maximum Combining (MC), Equal Gain Combining (EGC), and Maximum-Ratio Combining (MRC) are introduced based on multiple images obtained after repetitive transmission schemes. Besides, indoor OCC performance under various specific mobile phone camera orientations, user locations, and applied combining techniques are demonstrated. The second case is based on consecutive multiple images obtained without repetitive transmissions within a short time where practical mobile phone orientations induced by normal users' behavior are considered. Statistical performance analysis which includes descriptions of the transmitter imaging location and intensity distribution is provided. Besides, communication area coverage and supported independent transmitter numbers of the 2×2 LED transmitters at various user locations are studied considering receiver practical orientations during normal phone use. Our research aims to provide theoretical support for practical system design.

2. INDOOR OCC SYSTEM MODEL

We consider a typical 5 m × 5 m × 4 m indoor room as the OCC application scenario where 2 × 2 LED transmitters are installed on the room ceiling at the height of 3.95 m. These LEDs are based on On-off Keying (OOK) modulation at 500 Hz to carry location and identity-related information without human eyes' perception. We only consider downlinks in our study and users rely on their mobile phone cameras to acquire LED-delivered information to help determine their current locations. During the information acquisition process, the phone's background program can schedule the camera to capture the LEDs within its Field Of View (FOV) as needed to obtain intensity-carrying information without disturbing the users' normal phone use. As the phone orientations are arbitrary and only Line-Of-Sight (LOS) LED imaging is considered, there exist possibilities that none of the four LEDs can be imaged under which users fail to determine their current locations. Demonstrations of the indoor OCC system are shown in Figure 1. Some key parameters of the transceiver are listed in Table 1.

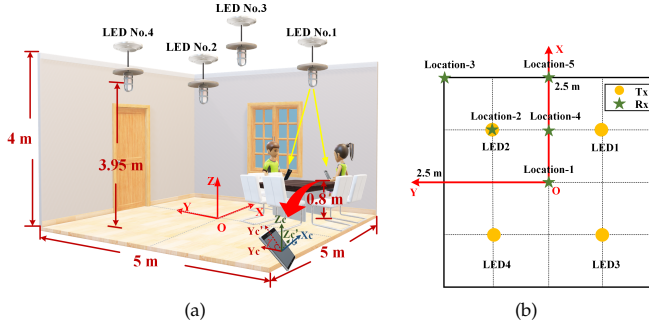


Fig. 1. Demonstrations of the indoor OCC system. (a) System layout. (b) Top view of four LED positions and some users' locations.

Table 1. Key Parameters of the Transceiver

Tx parameter	Value	Rx parameter	Value
Half-power angle	60°	FOV	90°
Power	100 mW	Resolution	1024 ²
Diameter	170 mm	Pixel size	6.5 μm

Imaging location and intensity are two aspects of the indoor OCC system model with the former helping to establish the OCC communication link and the latter benefiting the intensity-based information acquisition. Given the limited space and to highlight the focus of this study while maintaining its integrity, we directly provide related modeling methods and detailed derivations and accuracy validations can be referred to in our previously published papers [2] and [3]. For imaging location, given an arbitrary point (X_L, Y_L, Z_L) of LED under the World Coordinate System, its corresponding imaging point (u_L, v_L) can be calculated as

$$k \begin{bmatrix} u_L \\ v_L \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{p_u} & 0 & u_0 \\ 0 & \frac{1}{p_v} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R} & \mathbf{T} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} X_L \\ Y_L \\ Z_L \\ 1 \end{bmatrix}, \quad (1)$$

where k is the scale factor, f is the camera focal length, and (u_0, v_0) is the coordinate of the Image (Film) Coordinate System original point under the Pixel Coordinate System. p_u and p_v are the physical length of a pixel along the u -axis and v -axis respectively, and $\mathbf{T}$ is the translation matrix which describes the offset angle and $\mathbf{R}$ is the rotation matrix which describes the rotation angle. Matrix $\mathbf{R}$ can be calculated as

$$\mathbf{R} = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \beta & -\sin \beta \\ 0 & \sin \beta & \cos \beta \end{bmatrix} \begin{bmatrix} \cos \gamma & 0 & \sin \gamma \\ 0 & 1 & 0 \\ -\sin \gamma & 0 & \cos \gamma \end{bmatrix}, \quad (2)$$

where α , β , γ are the rotation angles of yaw, pitch, and roll which correspond to the rotation about the Z_c , X_c , and Y_c of the Receiver Coordinate System respectively. Demonstrations of the World Coordinate System (X, Y, Z) and Receiver Coordinate System (X_c, Y_c, Z_c) are shown in Figure 1(a), and only the receiver-induced angular distortions are considered. For imaging intensity, the final value of pixel (u_L, v_L) of an 8-bit Global Shutter (GS) camera is directly given as

$$PV_{(u_L, v_L)} = 255 \times \left\{ \frac{A_d A_c [V_{a_ref} - A_f (V_{ref} - Q_e A_n \frac{\int_{t_s}^{t_s+t_{exp}} E_i(u_L, v_L)(t) dt}{Q_p})]}{raw_{max}} \right\}^{1/g_a}, \quad (3)$$

where $E_{i(u_L, v_L)}$ is the irradiance of the (u_L, v_L) pixel which is influenced by the LED transmission power, irradiance angle, and channel Direct Current (DC) gain. t_s is the time point when the exposure of a frame starts, t_{exp} is the camera exposure time, Q_p and Q_e are the energy of a single photon and the quantum efficiency respectively. A_d , A_c , A_f , A_n are the Analog-to-Digital Converter (ADC) gain, Correlated Double Sampling (CDS) gain, source follower gain, and sense node gain. V_{a_ref} and V_{ref} are the maximum voltage that can be quantified by ADC and the reference voltage. raw_{max} is the maximum possible raw output value, and g_a is the camera gamma value.

Multiple images that our subsequent research relies upon are generated according to Eq.(1) and Eq.(3) under the following two cases' requirements in MATLAB. Specific generation methods can be found in our previous work [2, 3] where their accuracy and rationality are validated on our experimental testbed.

3. CASE 1: ANALYSIS BASED ON THREE IMAGE-DRIVEN COMBINING TECHNIQUES

A repetitive transmission scheme is a common solution to enhance communication reliability. In this case, we take multiple identical images acquired under the repeated transmission scheme as a whole study subject. To strike a balance between transmission efficiency and reliability, we assume the repeated transmission time is 3. Three image-driven combining techniques, including MC, EGC, and MRC, are applied to leverage these multiple images in suppressing random noise-induced error.

For MC, the maximum pixel value within each LED imaging area per image is determined first, and each LED imaging intensity is acquired by averaging the maximum pixel value within all studied images correspondingly. For EGC, the average pixel value within each LED imaging area per image is determined first, and each LED imaging intensity is acquired by averaging the average pixel value within all studied images correspondingly. For MRC, each LED imaging intensity is calculated as the weighted sum of the pixel values per pixel within the LED

imaging area. Each pixel weight coefficient is the ratio between the individual pixel Signal-to-Noise Ratio (SNR) and the sum of all pixel SNRs within each LED imaging area. Here, each pixel SNR is the ratio between the mean value of this pixel within all images and its mean square error.

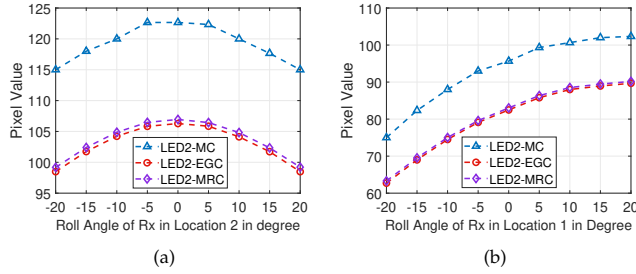


Fig. 2. LED2 imaging intensity variations with the camera roll angle orientations under MC, EGC, and MRC. (a) The user is at Location 2. (b) The user is at Location 1.

The user's position is first fixed and the LED imaging intensity variations with the camera angle orientations under MC, EGC, and MRC are provided in Figure 2 where we take LED2 as the example and two typical user locations are considered. For simplicity, only the camera's roll angle variations are included. The user is assumed to stand at Location 2 directly under LED2 in Figure 2(a), and LED2 imaging intensity under MC, EGC, and MRC are symmetrically distributed about the camera's roll angle due to the applied Lambertian light source. MC achieves the highest pixel values among the three techniques and the EGC is the lowest. However, when the user is assumed to stand at Location 1 which is the room center in Figure 2(b), LED2 imaging intensity under MC, EGC, and MRC all increase with the camera roll angle value due to the existence of both the location-induced offset angle and rotation angle. Moreover, pixel value differences between the three techniques are less obvious with the decreased LED imaging area compared between Figure 2(a) and (b).

As the user's location is seen to affect the imaging intensity distribution and their trend in Figure 2, LED imaging intensity variations with the user location variations under MC, EGC, and MRC are demonstrated further in detail in Figure 3(a) where five typical user locations are selected based on Figure 1(b) with no camera angle rotations existed. For LED2 imaging intensity in Figure 3(a), the MC-based values at each selected position are the highest and the EGC-based values at each selected position are the lowest. Pixel values achieved at Location 2 are slightly larger than those obtained at other locations due to the closest transceiver distance. Besides, as Locations 1, 3, and 5 are symmetrical relative to LED2, the pixel values obtained at these locations are the same. Moreover, intensity changes brought by location-induced offset are not as obvious as the rotation angle-led ones. We also provide four LED imaging intensity comparisons at different locations under MC in Figure 3(b). When the user is in the room centre, the four LED imaging intensities are the same due to the position symmetry. When the user is in either Location 2 or 4 where the four LEDs are within the camera FOV, their intensities vary according to different distances. When the user is in extreme positions such as Locations 3 and 5, some of the LEDs are out of view and cannot be imaged.

Communication performances of downlinks between the

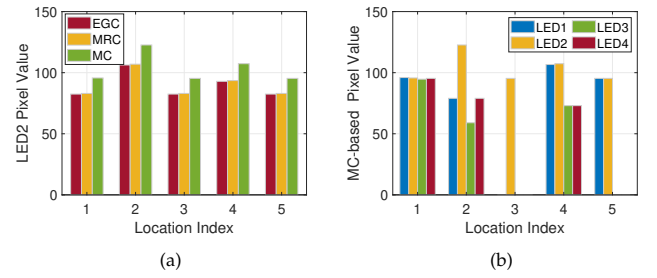


Fig. 3. LED imaging intensity variations at different user locations without camera rotations. (a) LED2 imaging intensity under MC, EGC, and MRC. (b) Four LEDs imaging intensity under MC.

user at Location 2 and LED2 under MC, EGC, and MRC are shown in Figure 4 where BER and channel capacity are provided. Among the three combining techniques, MC achieves the best in both BER and capacity aspects as it is more effective in combating random camera noise and supporting a wider range of imaging pixel values. MRC is superior to EGC in terms of BER and channel capacity as the weighted values introduced at each pixel comprehensively consider their different SNRs. Besides, performance under MC fluctuates less obviously than those of EGC and MRC.

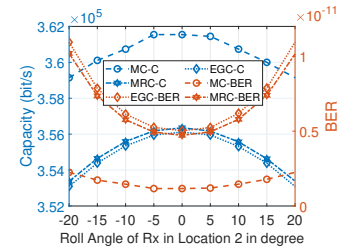


Fig. 4. Communication performances of downlinks between the user at Location 2 and LED2.

4. CASE 2: ANALYSIS CONSIDERING PRACTICAL RECEIVER CAMERA ORIENTATIONS

Most studies provide OCC performance analysis assuming the camera receiver is vertically upward and fixed. As the OCC channel is not isotropic and device orientation variations induced by normal users' behaviour are common, statistical OCC performance analysis considering the impact of random device orientations is necessary which is provided in this case.

Table 2. Normal phone use-induced receiver orientations

	α	β	γ
Mean	0	40.78	-0.84
Standard deviation	3.67	2.39	2.21
Coherence time	0.342	0.377	0.331

We only consider the sitting phone users and apply the measurement-based random process model for device orientation descriptions according to [7] where Laplace distributions

are validated to fit the measured data well. Statistics we use to describe the device random orientation are listed in Table 2 according to measurement data obtained in [7] and 1000 images are collected to realize the following statistical analysis. Probability density distributions of imaging location and pixel values based on MC and EGC are demonstrated in Figure 5 where images without LEDs are not included in the statistics. Gaussian and Laplacian fitting curves are also shown to provide a rough description of imaging location and pixel value probability density distributions. Moreover, the similarity of the obtained data with these two particular distributions is evaluated through Kolmogorov-Smirnov Distance (KSD) and kurtosis shown in Table 3.

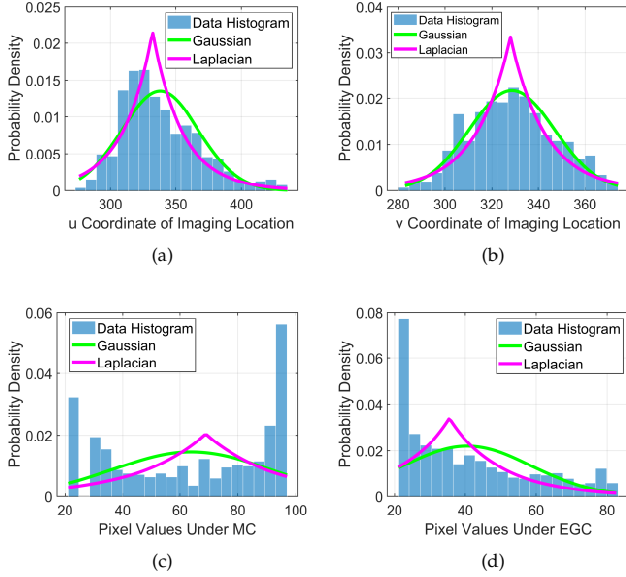


Fig. 5. Probability density distributions of imaging location and pixel values. (a) u coordinate. (b) v coordinate. (c) Pixel values under MC. (d) Pixel values under EGC.

Table 3. Statistics of imaging location and pixel values

	u	v	EGC	MC
Kurtosis	3.34	2.8	2.37	1.56
Gaussian KSD	0.091	0.0785	0.1411	0.1241
Laplace KSD	0.1417	0.1128	0.1788	0.1764

The KSD, which measures the distance between the acquired data and Gaussian or Laplace distributions (with kurtosis values of 3 and 6, respectively), indicates greater similarity with lower values. From Table 3, the kurtosis of u, v, and pixel values under EGC more closely match the Gaussian distribution, confirming that the imaging location and intensity are more similar to the Gaussian distribution than the Laplace distribution.

We analyze the support capacity for independent LED transmission numbers at various positions shown in Figure 6 with distributions for locations supporting one LED and four LEDs. The user X-Y plane is divided into a 5 cm grid with 100 points representing user locations. At each point, 500 images are captured, reflecting typical phone orientations during 1 second of

normal use. The number of images capturing 0, 1, 2, 3, and 4 LEDs is counted to assess transmission support. Most locations support at least one LED for transmission, except for a few edge positions. However, few locations allow simultaneous independent transmission of 4 LEDs where optimizing the LED layout is expected to expand their coverage.

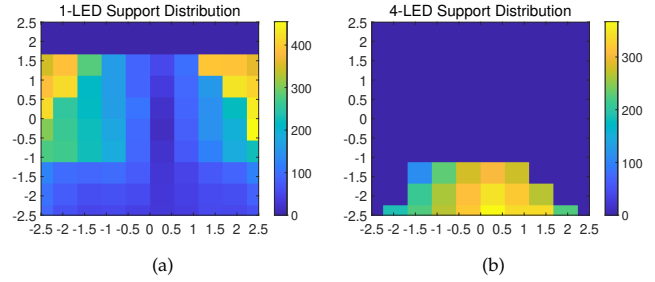


Fig. 6. Multi-LEDs communication supported location distribution. (a) One LED-supported. (b) Four LEDs-supported.

5. CONCLUSION

This Letter presents a statistical performance analysis of an indoor OCC system using multiple images through two cases. The first study evaluates the impact of three image-combining techniques on images obtained from repetitive transmissions. User location-induced angle offsets and phone orientations affect LED imaging intensity distributions. Among the techniques, MC demonstrates the best performance in terms of BER and capacity, while MRC outperforms EGC. The second study examines how user behavior-led random phone orientations impact imaging location and intensity where Gaussian distribution is shown to fit the data better than Laplace distribution. Besides, multi-LEDs communication supported location distribution is also provided. This study provides a statistical perspective on indoor OCC performance with plans to incorporate random user movement in future investigations.

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Data Availability Statement. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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